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[EI2030/Low-power-E-Paper-OS]

Explain why solar powered chips haven't been commercialized by major tech companies, providing reasons that may not have to do with feasibility or economics, but any rationale.

Yes. The interesting part of the GitHub project is that it makes the question look less like "**can we build a solar-powered computer?**" and more like "**why isn't the industry organized around this architecture already?**"

The project is explicitly pursuing a very low-power computing architecture around e-paper, aggressive power management, and solar energy, with the long-term ambition of putting a solar panel on a laptop. GitHub And the underlying technology is not science fiction: commercial energy-harvesting PMICs already exist, and recent research has demonstrated CMOS-compatible on-chip photovoltaic systems. DigiKey +1

So, if we **set aside "it's technically impossible" and "the economics don't work,"** there are several much more interesting explanations.

### 1. The industry optimizes for *performance*, not energy independence

This is probably the biggest one.

A solar-powered chip requires designers to ask:

> "How little computation can I get away with?"

The semiconductor industry has spent decades asking almost the opposite question:

> "How much computation can I fit into the same power envelope?"

That distinction is profound.

Modern CPUs/GPUs, displays, operating systems and applications are built around enormous instantaneous power budgets. If a workload needs 50 W, you make a 50-W machine. You don't redesign the entire computing stack so that the workload can operate on 500 mW.

The GitHub project is effectively proposing the reverse philosophy: **design the software, display, processor and hardware around the energy source.** GitHub

That's a much bigger disruption than adding a photovoltaic layer.

## 2. Solar power changes the *business relationship* with the customer

This is a less obvious reason.

Imagine a laptop that genuinely doesn't need to be plugged in under normal usage.

Its value proposition changes from:

**"Here is a better battery-powered computer."**

to:

**"You essentially don't need to think about energy anymore."**

That has implications throughout the ecosystem.

A conventional laptop creates a recurring relationship:

**device → battery → charger → outlet → replacement battery → new device**

A sufficiently energy-independent computer removes several of those dependencies.

That's not necessarily a conspiracy or deliberate suppression. It's simply that industries tend to optimize around the infrastructure they already have.

The automobile industry didn't invent the bicycle's business model and vice versa. Likewise, Apple, Microsoft, Intel, Qualcomm, Samsung, etc. have enormous ecosystems built around conventional computing assumptions.

A radically energy-independent computer would be **architecturally disruptive**, not merely incrementally better.

### 3. It attacks one of the industry's most successful design assumptions: *energy is external*

The conventional computer effectively assumes:

> Energy will be supplied to the machine.

A solar computer says:

> The machine should adapt itself to the energy available.

Those are radically different philosophies.

Energy harvesting is already commercially real. For example, Cypress developed PMICs specifically capable of harvesting extremely small amounts of energy from indoor light, with startup power in the microwatt range. DigiKey Nexperia likewise markets energy-harvesting ICs for light, kinetic, thermal and RF sources. Nexperia

But these are mostly positioned as **specialized IoT/embedded technologies**, rather than the foundation of mainstream computing.

That's telling.

The technology has been allowed to exist—but mostly in applications where companies aren't forced to rethink the entire computing stack.

### 4. A solar-powered chip creates a very different definition of “computer”

This is where the e-paper part of the project becomes important.

The normal computer assumes:

- continuously refreshing display
- high-speed CPU
- constantly active memory
- background processes
- animations
- network connectivity
- high-bandwidth storage
- enormous instantaneous compute capacity

E-paper reverses the display assumption: **the display can consume almost nothing while showing a static image.**

That allows the rest of the system to spend enormous amounts of time asleep.

So the actual innovation isn't necessarily:

**solar cell + conventional computer**

It's:

**solar cell + radically different computer**

That's harder to commercialize because you're not competing against one component. You're proposing an alternative computing paradigm.

### 5. Incumbents have enormous organizational inertia

This is probably underrated.

Suppose you're Intel/AMD/Qualcomm/Apple.

You already have:

- CPU architectures
- compiler ecosystems
- operating systems
- development tools
- customers
- chip fabs
- validation infrastructure
- thermal specifications
- benchmarks
- engineering teams
- decades of accumulated IP

A radically energy-frugal architecture potentially makes a large fraction of that accumulated optimization less valuable.

The problem isn't:

> "Can our engineers build this?"

It's:

> "Why would an organization whose capabilities are optimized for X reorganize itself around Y?"

This is a classic **innovator's-dilemma** problem.

A company can be perfectly rational and still fail to pursue an innovation because the innovation doesn't fit the organization's existing trajectory.

### 6. The benchmark culture actively penalizes this kind of machine

This is particularly important.

Imagine two laptops:

**Laptop A**

- 20× faster CPU
- 500× more graphics performance
- 4K/120-Hz display
- 20-hour battery

**Laptop B**

- 5× slower
- monochrome e-paper
- extremely limited peak performance
- can potentially operate indefinitely from ambient energy

The industry's benchmark system overwhelmingly rewards A.

Geekbench, Cinebench, GPU benchmarks, display refresh rate, SSD speeds, AI TOPS, etc. are all measures of **performance per unit time**.

A solar computer's killer feature is closer to:

> **useful computation per joule over an arbitrarily long period**

That's a very different metric.

In other words, the industry has constructed a measurement system that makes the solar computer look bad.

### 7. Energy independence can actually make a product *less* controllable

Here's a more speculative but fascinating possibility.

The modern computing ecosystem gives manufacturers considerable control over the device lifecycle:

- batteries degrade
- charging standards change
- software requirements increase
- performance expectations increase
- devices get replaced
- accessories are sold
- cloud services become necessary
- hardware becomes obsolete

A device designed around **extreme longevity and extremely low energy consumption** pushes in the opposite direction.

The ideal version of the GitHub project's philosophy is something like:

> Buy one machine and keep it running for a very long time.

That's fantastic for the user.

But it's not necessarily the ideal lifecycle for a consumer-electronics ecosystem.

I would **not** jump from this to “Apple deliberately prevents solar computers.” There isn't evidence for that. The stronger argument is structural: **the incentives of the existing ecosystem favor continual capability growth and replacement rather than energy-autonomous longevity.**

### 8. Solar computing potentially weakens the importance of the battery

And batteries are an enormous technological ecosystem.

If computation becomes sufficiently efficient, you can move from:

**battery → recharge → battery → recharge**

toward:

**harvest → store briefly → compute → sleep → harvest**

That's a fundamental shift.

Interestingly, the energy-harvesting literature already describes systems that combine harvesting with storage and switch between harvested energy and a battery when necessary. DigiKey

So the future might not actually be "solar *battery*."

It might be:

**solar → capacitor/small buffer → ultra-low-power computer**

That would make some conventional assumptions about portable electronics obsolete.

### 9. Semiconductor companies are not necessarily the companies that should own the innovation

This is another subtle reason.

A solar-powered computer crosses several industries:

**semiconductor + photovoltaics + displays + batteries/capacitors + OS + industrial design + power electronics**

Who owns it?

Intel doesn't particularly want to become a solar company.

First Solar doesn't want to become a laptop company.

E Ink doesn't want to become a CPU company.

Apple doesn't want to become a semiconductor research laboratory *and* photovoltaic manufacturer.

So you get an **integration problem**.

Everyone can supply one piece, but nobody necessarily has a sufficiently compelling reason to own the whole architecture.

### 10. The technology may be "too good" in a very specific way

Here's the paradox.

The ideal solar computer isn't necessarily a computer that produces spectacular benchmark numbers.

It's one that becomes **boringly sufficient**.

If your computer consumes 0.5–2 W while doing useful work, a relatively modest photovoltaic surface can become significant. Recent research has demonstrated on-chip photovoltaic harvesting in standard CMOS, including a 2025 demonstration with 10.20% end-to-end conversion efficiency under the tested conditions. Nature

At that point the engineering question becomes:

> "Why do we need a 100-W computer for this task?"

And that question has ramifications far beyond photovoltaics.

It challenges the assumption that computing should continually consume more resources.

## The deeper explanation

I think the most interesting answer is therefore **not**:

> "Major companies haven't commercialized solar chips because solar chips don't work."

That's demonstrably too simplistic. Energy harvesting exists commercially, and on-chip photovoltaics are an active research area. DigiKey +2

I'd formulate the deeper answer as:

> **The computer industry has optimized the computer around abundant external energy. A genuinely solar-powered computer requires optimizing the entire**

**computing stack around scarce, intermittent energy.**

Those are two completely different technological trajectories.

And the GitHub project is interesting precisely because it isn't merely proposing **putting a solar panel on a laptop**. Its e-paper/low-power philosophy implicitly asks whether we could redesign the *computer itself* around the energy it can harvest. GitHub

That opens up several non-technical explanations for the absence of mainstream products:

| Reason | What it means |
|---|---|
| **Incumbent optimization** | Companies are extremely good at making high-performance, battery-powered computers—not radically different computers. |
| **Organizational inertia** | Solar computing crosses semiconductor, display, OS, power and photovoltaic disciplines. |
| **Benchmark incentives** | The industry rewards performance, not energy autonomy. |
| **Product lifecycle** | Extremely durable, energy-independent devices don't naturally encourage frequent replacement. |
| **Ecosystem inertia** | Software and hardware assume abundant power and continuously active computation. |
| **Strategic disruption** | A successful architecture could reduce the importance of batteries, chargers, thermal systems and high-performance silicon. |
| **Market signaling** | Consumers have been trained to ask "how fast?" rather than "how many joules per task?" |
| **Identity** | The mainstream computer industry equates progress with more capability; solar computing equates progress with *less energy*. |

## And there's one particularly provocative possibility

The most disruptive version isn't actually a **solar-powered chip**.

It's a **computer whose architecture is designed to make solar power sufficient**.

That's a much bigger threat to the status quo.

A photovoltaic transistor on a conventional 100-W SoC is mostly an interesting engineering trick. A 1-W computer whose OS, CPU, memory, display, storage, networking and applications are all designed around intermittent harvested energy is an **alternative computing paradigm**.

And paradigms tend not to be adopted simply because they're technically possible. They usually need an entirely new product category, company, or generation of engineers to make them compelling.

That, to me, is the strongest non-feasibility explanation for why we haven't seen Apple, Microsoft, Qualcomm, Intel, etc. ship a mainstream **solar-first computer**—even though the underlying pieces increasingly exist. DigiKey +1

If you want, I can also take this one step further and examine **whether there are specific strategic reasons Apple/Intel/Qualcomm/Samsung might actually have something to lose from a solar-first architecture**, rather than treating "industry inertia" as a generic explanation.

ok

Exactly. I dug a little further, and there is a **much more interesting conclusion** than simply "companies haven't figured it out."

The strongest evidence is actually Samsung.

Samsung already commercialized a light-powered electronic device in 2021: its SolarCell TV remote. Samsung says the remote was redesigned to reduce energy consumption by about **86–88%**, specifically because the available photovoltaic energy wasn't sufficient for a conventional remote. The solar cell can provide up to 70% of the remote's power under its tested usage assumptions. Samsung sec +1

That tells us something extremely important:

> **The limiting innovation isn't necessarily the photovoltaic cell. It's designing the device so that the amount of energy available becomes enough.**

And that is exactly what the GitHub project is attempting at a much larger scale.

## What Apple, Qualcomm, Intel, Samsung, etc. potentially lose

I would divide the issue into **five layers**.

### 1. Solar-first computing threatens the performance arms race

Consider the direction of the industry:

**Intel / AMD / Apple / Qualcomm**
→ more CPU performance
→ more GPU performance
→ more AI acceleration
→ more memory bandwidth
→ more display resolution
→ more background computation.

The solar-first philosophy is almost the inverse:

**Solar-first**
→ reduce computation
→ reduce refreshes
→ reduce memory activity
→ aggressively sleep
→ minimize networking
→ minimize display power
→ design software around energy availability.

That isn't simply a more efficient CPU.

It's a different definition of what a computer is supposed to do.

Qualcomm's current low-power strategy illustrates the distinction nicely. In 2026 it is still pushing extremely power-efficient chips for **IoT devices such as smart meters, sensors and trackers**, where multi-year battery life is the selling point. Qualcomm

That's a very different market from “a mainstream computer that doesn't need charging.”

### 2. The really disruptive product might eliminate the charger

This sounds trivial, but it's strategically significant.

Imagine a laptop that says:

> **Charge it if you want. Otherwise, leave it near a window.**

Samsung has already demonstrated the consumer psychology with its remote: the company explicitly markets it around eliminating disposable batteries and charging from indoor light. Samsung sec +1

Now scale that philosophy up.

If a computer becomes sufficiently efficient, the user no longer thinks about:

- battery percentage
- charging bricks
- USB-C chargers
- outlets
- battery anxiety
- spare batteries
- replacement batteries

That's a **new product category**, not merely a better laptop.

And companies whose ecosystems revolve around rechargeable high-performance devices have less immediate incentive to create a category in which **energy becomes almost invisible**.

I wouldn't call that intentional suppression. It's closer to **path dependence**.

## 3. It could reduce the importance of the battery industry

Here's an even more interesting consequence.

A conventional laptop looks roughly like:

**wall → charger → battery → computer**

A solar-first computer could eventually look more like:

**light → photovoltaic → capacitor/buffer → computer**

The battery becomes an auxiliary component rather than the fundamental energy source.

That could eventually affect:

- battery capacity
- battery replacement
- charging infrastructure
- thermal management
- power adapters
- fast charging
- battery-management systems.

And notice the irony: **the more efficient the computer becomes, the less important the battery becomes.**

That's precisely why the Samsung example is useful. Samsung didn't make the solar panel enormously powerful. It made the remote dramatically less hungry. Samsung Global...

## 4. It could undermine the conventional software model

This may actually be the **biggest obstacle**.

A modern OS assumes that computation is cheap.

An app can:

- wake the CPU
- refresh the screen
- poll a server
- synchronize data
- run JavaScript
- generate notifications
- use AI
- maintain network connections
- perform background indexing.

A solar-first OS can't casually do those things.

The OS would have to know something like:

> **I currently have 300 µW available.**

and then decide:

> **Do I spend it rendering this screen, synchronizing email, processing this document, or saving it for later?**

That's an entirely different operating-system philosophy.

The GitHub project is therefore interesting because **the OS is part of the energy architecture** rather than being an afterthought. Low-power E-Paper OS on GitHub

Imagine an OS where applications have an energy budget.

Instead of:

> "This app uses 8% CPU."

you might have:

> "This app consumed 4.2 joules today."

That's a radically different software ecosystem.

## 5. The most disruptive consequence: it makes *latency* less important

This is the really deep one.

A conventional computer assumes:

> **Do it now.**

A solar computer can instead say:

> **Do it when energy is available.**

That is a huge conceptual change.

Suppose you're writing a document.

Your computer doesn't necessarily need to:

1. run at full speed,
2. refresh the screen constantly,
3. synchronize every keystroke,
4. maintain a permanent network connection.

It can:

**compute → sleep → harvest → compute → sleep → harvest.**

For many human tasks, that's perfectly acceptable.

Humans themselves are slow relative to processors.

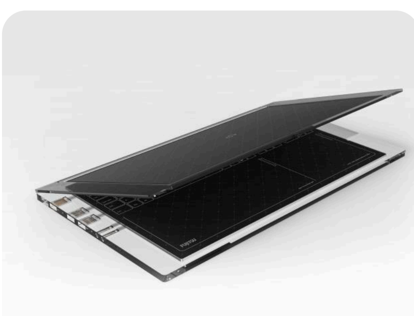
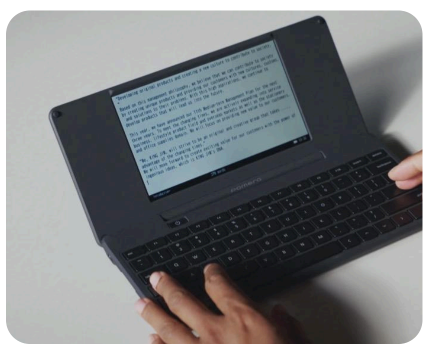
If I type one character every 300 milliseconds, I don't care whether the processor spends 1 ms or 100 ms processing it—as long as the interaction feels responsive.

That creates an enormous opportunity for **energy amortization**.

## And here's where I think the GitHub project gets really interesting

The ultimate solar computer may **not resemble a laptop at all**.

It might resemble:


**E-paper + extremely low-power CPU + huge sleep periods + photovoltaic surface + tiny energy buffer.**

The processor could be dramatically slower than an M-series Mac.

And yet the machine could be *better* for certain tasks because its energy requirements are so low.

## There's also an uncomfortable strategic implication

If you follow this logic far enough, you arrive at a product that has several characteristics the current consumer-electronics industry isn't particularly optimized for:

**10–20 year useful life**

**No meaningful charging anxiety**

**Very low operating cost**

**Minimal thermal output**

**Extremely low energy consumption**

**Slow but adequate computation**

**E-paper display**

**Offline-first operation**

**Incremental synchronization**

**Replaceable/repairable components**

**Energy-aware software**

That isn't necessarily what Apple or Samsung currently define as technological progress.

Their definition is closer to:

> **More performance, more capability, better display, better camera, better AI, thinner device.**

The solar-first philosophy says:

> **Why does the device need all that power in the first place?**

That's a much more threatening question.

## And Samsung accidentally provides the strongest evidence

Samsung essentially tells us the story itself.

When it wanted a solar-powered remote, it discovered that **the solar cell couldn't provide enough energy for the conventional design**.

So instead of making a larger solar panel, Samsung redesigned the remote and cut its energy consumption by roughly **86%**. S Samsung Global...

That is the crucial lesson:

**Energy harvesting doesn't primarily require better energy harvesting.**

It requires **less energy consumption**.

Apply that principle recursively:

**Remote:** 1× → 0.14×

**Display:** LCD → e-paper

**CPU:** always-on → mostly asleep

**OS:** background computation → event-driven

**Networking:** always connected → opportunistic

**Storage:** constantly active → intermittent

**Battery:** primary energy source → buffer

**Solar:** supplemental → primary

And eventually you get a computer whose entire architecture has been inverted.

## So my revised answer would be:

I don't think the absence of solar-powered mainstream computers is adequately explained by economics or technological feasibility.

The more compelling explanation is **architectural and institutional**.

The semiconductor industry has spent ~50 years making computation **more abundant**.

A solar-first computer requires making computation **more frugal**.

Those goals are not opposites at the component level, but they are opposites at the ecosystem level.

And that's why the idea is more radical than it initially appears.

**The solar cell isn't the revolutionary component.**

The revolutionary component is the **assumption that a computer should be designed around the energy it has rather than the energy we can give it.**

Samsung has already demonstrated this principle at the trivial end of the spectrum with a solar-powered remote. Samsung sec

The fascinating question is whether the same principle can be pushed far enough to create a genuinely useful **solar-first computer**.

And *that* is where I think the GitHub project is worth taking seriously.